

Nicola De Carli

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Research Experience

- Feb 2025 – present **Postdoctoral Researcher with Dimos Dimarogonas**, KTH Royal Institute of Technology, Sweden
- May 2024 – Jan 2025 **Postdoctoral Researcher with Paolo Robuffo Giordano, Marco Tognon**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- June 2022 – Dec 2022 **Visiting Ph.D. Student**, Centro Piaggio, University of Pisa, Italy
- Jan 2021 – Apr 2024 **Ph.D. Student**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Title of the thesis: Active Perception and Localization for Multi-Robot Systems
 - Keywords: Multi-Robot Systems, Active Perception, Localization, Constrained Control
 - Time for the preparation of the thesis: 3 years and 4 months (01 January 2021 - 17 April 2024)
 - Supervisors: Paolo Robuffo Giordano, Paolo Salaris
 - Jury: Dimos Dimarogonas, Eduardo Montijano, Isabelle Fantoni, Fabio Morbidi

Education

- Jan 2021 – Apr 2024 **IRISA-CNRS, Université de Rennes**, Ph.D. in Automation, Production and Robotics Engineering
- Title of the thesis: Active Perception and Localization for Multi-Robot Systems
- Sept 2019 – Aug 2020 **Ecole centrale de Nantes**, MS-Erasmus in Robotics Engineering
- Sept 2018 – Oct 2020 **Università degli Studi di Genova**, MS in Robotics Engineering
- Title of the thesis: Formation Maintenance of Quadrotor Swarms Using Second Order Visual Control
- Sept 2015 – Oct 2018 **Università degli Studi di Genova**, BS in Electronics and Communication Engineering
- Title of the thesis: Implementation of Computer Vision Algorithms on Embedded Systems

Ph.D. Student Supervision

- Feb 2025 – present **Co-supervisor**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Valeria Braglia, Ph.D. Thesis student on "Perception-Aware Cooperative Cable Suspended Transportation"
 - Co-supervised with Marco Tognon and Paolo Robuffo Giordano
- Sept 2024 – present **Co-supervisor**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Paul Mefflet, Ph.D. Thesis student on "Shared Telemanipulation"
 - Co-supervised with Marco Ferro, Claudio Pacchierotti, Marco Aggaravi and Paolo Robuffo Giordano

Master's Student Supervision

- Feb 2026 – Aug 2026 **Co-supervisor**, IRISA-CNRS, INRIA Rennes - Bretagne Atlantique, France
- Federico Calzoni, Master Thesis student on "Distributed Model Predictive Control for Cable Suspended Transportation"

- Co-supervised with Valeria Braglia, Marco Tognon, Andrea Drudi and Giuseppe Notarstefano
- Jan 2026 – June 2026 **Supervisor**, KTH Royal Institute of Technology, Sweden
- Héctor Adriel Sanvicente Solís, Master Thesis student on "Game-Theoretic Receding Horizon Planner for Underwater Reach-Avoid Games"
 - Co-supervised with Emilio Benenati
- Jan 2026 – June 2026 **Supervisor**, KTH Royal Institute of Technology, Sweden
- João Miguel Gonçalves Zenário, Master Thesis student on "Sensor-based Formation Control of Underwater Multi-Robot Systems Using Barrier Lyapunov Functions"
 - Co-supervised with Victor Nan Fernandez-Ayala
- Sept 2023 – Feb 2024 **Co-supervisor**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Riccardo Belletti, Master Thesis student on "Distributed NMPC-based Full-Pose Control of a Cable Suspended Load Using Multiple UAVs"
 - Co-supervised with Marco Tognon
- Feb 2024 – July 2024 **Co-supervisor**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Emanuele Buzzurro, Master Thesis student on "Distributed NMPC-based Full-Pose Control of a Cable Suspended Load Using Multiple UAVs"
 - Co-supervised with Marco Tognon and Esteban Restrepo
- Mar 2024 – Aug 2024 **Co-supervisor**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Valeria Braglia, Master Thesis student on "Perception-Aware Cooperative Cable Suspended Transportation"
 - Co-supervised with Marco Tognon and Esteban Restrepo
- Sept 2022 – Feb 2023 **Co-supervisor**, IRISA - CNRS, INRIA Rennes - Bretagne Atlantique, France
- Lorenzo Balandi, Master Thesis student on "Persistent Monitoring of Multiple Moving Targets Using High Order Control Barrier Functions"
 - Co-supervised with Paolo Robuffo Giordano

Invited Talks

- Oct 2024 **Talk at NYU Tandon School of Engineering**
- Title: Advances in Cooperative Cable-Suspended Transportation: Perception-Aware and Decentralized Control
- Oct 2024 **Talk at KTH Royal Institute of Technology, Division of Decision and Control Systems**
- Title: Advances in Cooperative Cable-Suspended Transportation: Perception-Aware and Decentralized Control

Conference Talks

- June 2025 **2025-IEEE Conference on Decision and Control**
- Title: A Distributed Kalman-like Observer with Dynamic Inversion-Based Correction for Multi-Agent Estimation
- June 2025 **2025-Reglermöte (Swedish Control Conference)**
- Title: A Distributed Kalman-like Observer with Dynamic Inversion-Based Correction for Multi-Agent Estimation
- Dec 2024 **2024-IEEE Conference on Decision and Control**
- Title: Adaptive Observer From Body-Frame Relative Position Measurements for Cooperative Localization
- May 2024 **2024-IEEE International Conference on Robotics and Automation**
- Title: Distributed Control Barrier Functions for Global Connectivity Maintenance

- Dec 2023 **2023-IEEE International Symposium on Multi-Robot and Multi-Agent Systems**
 - Title: Multi-Robot Active Sensing for Bearing Formations
- Dec 2022 **Journée GT2 UAV GDR in LS2N, Nantes, France**
 - Title: Active Cooperative Localization of Multi-Drone Systems using Distributed Optimization
- Aug 2022 **MRS Summer School, Prague, Czech Republic**
 - Title: Active Cooperative Localization of Multi-Drone Systems using Distributed Optimization
- Dec 2021 **Journée GT2 UAV GDR (Online)**
 - Title: Online Decentralized Perception-Aware Path Planning for Multi-Robot Systems
- Nov 2021 **2021-IEEE International Symposium on Multi-Robot and Multi-Agent Systems**
 - Title: Online Decentralized Perception-Aware Path Planning for Multi-Robot Systems

Other Activities

Associate Editor for IEEE Robotics and Automation Letters (RA-L) 2026

Associate Editor for IEEE International Conference on Intelligent Robots and Systems (IROS) 2026

Associate Editor for IEEE International Conference on Intelligent Robots and Systems (IROS) 2025

Languages

Languages: English (fluent), French (good), Italian (mother tongue)

Journal Publications

- De Carli, N., Belletti, R., Buzzurro, E., Testa, A., Notarstefano, G., & Tognon, M. (2025). Distributed NMPC for Cooperative Aerial Manipulation of Cable-Suspended Loads. IEEE Robotics and Automation Letters. [PDF](#)
- Pagano, F., De Carli, N., Restrepo, E., Marino, A., & Robuffo Giordano, P. (2025). Distributed Multi-Robot Active-Sensing of a Diffusive Source. IEEE Robotics and Automation Letters. [PDF](#) [Video](#)
- De Carli, N., & Dimarogonas, D. V. (2025). A Distributed Kalman-like Observer with Dynamic Inversion-Based Correction for Multi-Agent Estimation. IEEE Control Systems Letters. [PDF](#) [Code](#)
- De Carli, N., Restrepo, E., & Robuffo Giordano, P. (2024). Adaptive Observer From Body-Frame Relative Position Measurements for Cooperative Localization. IEEE Control Systems Letters. [PDF](#)
- Balandi, L., De Carli, N., & Robuffo Giordano, P. (2023). Persistent Monitoring of Multiple Moving Targets Using High Order Control Barrier Functions. IEEE Robotics and Automation Letters. [PDF](#) [Video](#)

Conference Publications

- Braglia, V., De Carli, N., Restrepo, E., & Tognon, M. (2026, September). Perception-Aware Control for Cooperative Cable-Suspended Transportation Under Sensing Constraints. In IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). [PDF](#)

- Jang, I., Marchesini, G., De Carli, N., Kim, B., Hwang, S., Kim, D., Krantz, E., Kong, Y., Jiang, F. J., Wong, A., Roque, P., Sanjaya, P. W. L., Bastianello, N., Dhullipalla, M. H., Johansson, K. H., Shim, H., Dimarogonas, D. V., & Kim, H. J. (2026). Demonstration of Space Robot Teleoperation over a Lossy and Delayed Network using ATMOS. In Proceedings of the 23rd IFAC World Congress. [PDF](#) [Video](#)
- Verhagen, J., Krantz, E., Sidrane, C., Dörner, D., De Carli, N., Roque, P., Mao, H., Tibert, G., Stenius, I., Fuglesang, C., Dimarogonas, D. V., & Tumova, J. (2026). Validation of Space Robotics in Underwater Environments via Disturbance Robustness Equivalency. In IEEE International Conference on Robotics and Automation (ICRA). [PDF](#) [Code Project](#)
- De Carli, N., Salaris, P., & Robuffo Giordano, P. (2024, May). Distributed Control Barrier Functions for Global Connectivity Maintenance. In IEEE International Conference on Robotics and Automation (ICRA) (pp. 12048–12054). [PDF](#) [Video](#) [Code](#)
- De Carli, N., Salaris, P., & Robuffo Giordano, P. (2023, December). Multi-Robot Active Sensing for Bearing Formations. In 2023 International Symposium on Multi-Robot and Multi-Agent Systems (MRS) (pp. 184–190). IEEE. [PDF](#)
- De Carli, N., Salaris, P., & Robuffo Giordano, P. (2021, November). Online Decentralized Perception-Aware Path Planning for Multi-Robot Systems. In 2021 International Symposium on Multi-Robot and Multi-Agent Systems (MRS) (pp. 128–136). IEEE. [PDF](#)

Under Review

- De Carli, N., Zenário, J., Fernandez-Ayala, V. N., & Dimarogonas, D. V. (2026). Vision-based Underwater Formation Control With Input Saturations via Barrier Lyapunov Functions. Under review for IEEE International Conference on Robotics and Automation (ICRA 2027). [PDF](#) [Code](#) [Video](#)
- Şekercioğlu, P., De Carli, N., & Dimarogonas, D. V. (2026). Safe Formation Control of Open Multi-Robot Systems with Connectivity-Preserving Reconfiguration. Under review for IEEE International Conference on Robotics and Automation (ICRA 2027). [PDF](#) [Code](#) [Video](#)
- Marchesini, G., De Carli, N., Cho, S., Kong, Y., Krantz, E., Dhullipalla, M. H., Dimarogonas, D. V., & Kim, H. J. (2026). Multi-Agent Transportation of Free-Flyers in Microgravity Via Pushing Interaction Under Human-in-the-Loop Control. Under review for IEEE International Conference on Robotics and Automation (ICRA 2027). [PDF](#) [Video](#)
- De Carli, N., Marchesini, G., & Dimarogonas, D. V. (2026). STL-GCS A Planner-Controller Framework for Signal Temporal Logic via Graphs of Time-varying Convex Sets. Under review for IEEE Transactions on Robotics. [PDF](#)
- De Carli, N., Bastianello, N., & Dimarogonas, D. V. (2026). ADMM-Based Distributed Kalman-like Observer with Applications to Cooperative Localization. Under review for IEEE Transactions on Automatic Control. [PDF](#) [Code](#)
- Torroba, I., Dörner, D., Fernandez-Ayala, V. N., Kartasev, M., Verhagen, J., Krantz, E., Marchesini, G., Ljung, C., Roque, P., Sidrane, C., Van der Spaa, L., De Carli, N., Ögren, P., Fuglesang, C., Tumova, J., Dimarogonas, D. V., & Stenius, I. (2026). Marinarium a New Arena to Bring Maritime Robotics Closer to Shore. Under review for IEEE Transactions on Field Robotics. [PDF](#)

Thesis

- De Carli, N. (2024). Active Perception and Localization for Multi-Robot Systems. Ph.D. Thesis, IRISA-CNRS, Université de Rennes, France. [PDF](#)